

SOUTH UIST RANGE, United Kingdom

WMO: 030230

Lat: 57.3575N

Lon: 7.3850W

Elev: 4

StdP: 101.28

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.4	0.6	-4.3	2.6	2.3	-3.1	2.9	2.5	22.2	7.4	20.3	6.7	3.8	10	0.875

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.2	19.4	16.0	17.8	15.1	16.7	14.5	16.7	18.5	15.8	17.1	15.1	16.2	6.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.3	17.5	15.1	10.7	16.4	14.5	10.3	15.7	47.0	18.8	44.3	17.2	42.2	16.3	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	17.0	15.0	-2.1	22.5	1.2	1.8	-3.0	23.8	-3.7	24.8	-4.4	25.9	-5.3	27.2
			-2.7	18.2	1.3	0.9	-3.7	18.8	-4.4	19.4	-5.1	19.9	-6.1	20.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	5.7	5.7	6.4	8.0	10.2	12.2	13.8	13.9	12.7	10.3	7.7	6.1	(6)
(7)		DBStd	3.71	2.18	2.28	2.11	2.26	2.25	1.83	1.67	1.57	1.71	2.19	2.48	2.60	(7)
(8)		HDD10.0	677	132	122	112	65	24	2	0	0	2	23	74	120	(8)
(9)		HDD18.3	3255	391	354	369	309	253	184	141	137	169	249	319	378	(9)
(10)		CDD10.0	463	0	0	1	6	29	68	118	122	82	32	5	1	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	1	0	0	0	0	(11)
(12)		CDH23.3	1	0	0	0	0	0	0	1	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.7	8.9	8.7	8.3	7.4	7.0	6.7	6.2	6.6	7.7	8.1	8.5	8.7	(14)
(15)	Precipitation	PrecAvg	1284	153	109	106	75	63	71	82	99	107	137	138	136	(15)
(16)		PrecMax	1694	262	254	215	122	159	132	142	265	208	233	225	240	(16)
(17)		PrecMin	940	65	46	20	31	5	22	35	42	32	52	91	30	(17)
(18)		PrecStd	171	48	51	41	23	35	27	25	46	41	54	31	53	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.2	13.0	17.1	20.2	20.6	22.0	21.5	18.6	16.0	14.1	12.1	(19)
(20)			MCWB	10.0	9.5	10.0	11.3	15.4	16.3	17.3	18.3	14.8	13.9	12.2	11.3	(20)
(21)		2%	DB	10.2	10.1	10.8	13.5	17.1	17.7	19.1	18.7	16.7	14.7	12.6	11.1	(21)
(22)			MCWB	9.4	9.2	9.1	10.3	13.4	14.7	16.3	16.4	14.3	13.1	11.3	10.3	(22)
(23)		5%	DB	9.6	9.6	9.9	12.0	15.0	16.1	17.5	17.2	15.8	14.0	11.8	10.5	(23)
(24)			MCWB	8.9	8.8	8.8	9.8	12.0	13.7	15.3	15.4	14.0	12.5	10.6	9.7	(24)
(25)		10%	DB	9.0	9.0	9.4	11.0	13.4	14.8	16.4	16.2	15.1	13.3	11.2	9.9	(25)
(26)			MCWB	8.2	8.1	8.3	9.3	11.1	12.9	14.6	14.5	13.7	11.9	9.9	9.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.2	10.5	12.1	15.7	16.7	18.1	18.6	16.0	14.8	13.0	11.6	(27)
(28)			MCDWB	10.9	10.5	11.8	15.2	19.1	19.6	21.3	20.3	17.1	15.2	13.3	11.9	(28)
(29)		2%	WB	9.8	9.7	9.7	11.0	14.0	15.2	16.7	16.8	15.2	13.7	11.7	10.6	(29)
(30)			MCDWB	10.0	10.0	10.3	12.4	16.4	17.2	18.5	18.3	15.9	14.3	12.2	11.0	(30)
(31)		5%	WB	9.2	9.1	9.2	10.3	12.6	14.1	15.7	15.7	14.7	13.0	11.0	10.0	(31)
(32)			MCDWB	9.6	9.5	9.7	11.5	14.3	15.5	17.1	16.8	15.4	13.7	11.6	10.5	(32)
(33)		10%	WB	8.4	8.4	8.6	9.7	11.6	13.4	14.9	15.0	14.1	12.3	10.2	9.2	(33)
(34)			MCDWB	8.9	8.9	9.3	10.7	13.0	14.6	16.1	15.8	14.9	13.1	11.0	9.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.6	3.8	4.1	4.4	4.7	4.2	4.1	4.2	3.9	3.9	3.8	3.9	(35)
(36)			MCDBR	3.5	3.3	4.0	5.5	7.0	6.0	5.9	5.2	4.4	3.9	3.6	3.8	(36)
(37)			MCWBR	4.0	3.4	3.5	3.8	4.5	3.8	3.8	3.8	3.4	3.5	3.9	4.3	(37)
(38)		5% WB	MCDBR	3.6	3.2	3.8	4.8	6.1	5.1	5.4	4.8	3.8	3.6	3.8	3.9	(38)
(39)			MCWBR	4.1	3.5	3.6	3.8	4.5	3.7	3.7	3.9	3.6	3.8	4.4	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.355	0.361	0.376	0.387	0.388	0.376	0.379	0.370	0.372	0.354	0.339	0.332	(40)	
(41)		taud	2.126	2.234	2.283	2.313	2.365	2.445	2.449	2.480	2.431	2.356	2.202	2.110	(41)	
(42)		Ebn at Noon	464	662	767	824	848	864	853	837	772	670	487	377	(42)	
(43)		Edh at Noon	59	79	97	110	111	104	102	92	84	69	53	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.37	0.98	2.15	3.73	4.93	5.02	4.60	3.66	2.40	1.25	0.50	0.25	(44)	
(45)		RadStd	0.04	0.10	0.16	0.26	0.37	0.50	0.32	0.24	0.12	0.12	0.04	0.03	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page